1. Aim of the practical: **a)** WAP that applies the decision tree on plant dataset.

b) WAP that applies the naive Bayes algorithm on data set.

2. Objective:

a) To apply the Decision Tree classification algorithm on the Iris plant dataset, visualize the decision rules, and evaluate the model performance in classifying plant species.

b) To implement the Naïve Bayes classification algorithm on a given dataset, analyze how probabilistic classification works, and evaluate the accuracy of the model.

3. Algorithm/Flowchart:

A.

```
import numpy as np
import pandas as pd
df=pd.read_csv("sample_dataset.csv")
df
```

```
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree
x=df.iloc[:,0:4]
y=df.iloc[:,-1]
from sklearn.model_selection import train_test_split
x = pd.get_dummies(x)
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3)
decision_tree = DecisionTreeClassifier(criterion='entropy', max_depth=4)
decision_tree.fit(x_train, y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy', max_depth=4)
```

```
decision_tree.fit(x,y)
y_pre=decision_tree.predict(x_test)
y_pre
import matplotlib.pyplot as plt
tree.plot_tree(decision_tree,filled=True)
decision_tree.score(x_test,y_test)
```

B.

```
from sklearn.datasets import load_iris
iris=load_iris()
```

```
x=iris.data
y=iris.target
```

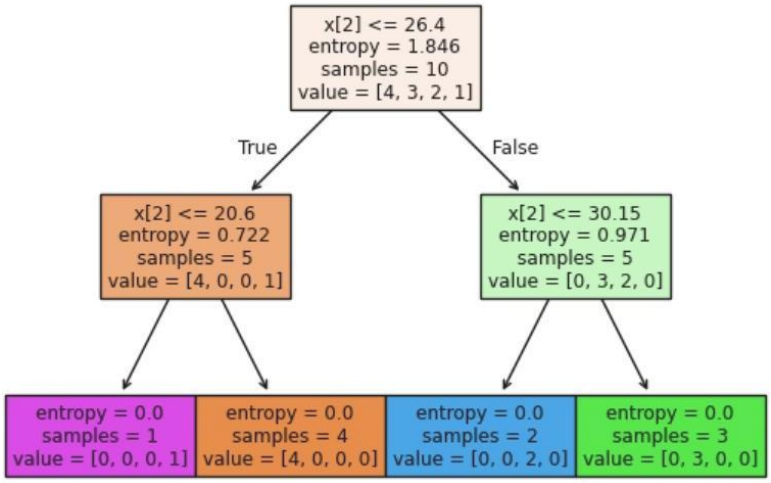
```
print(iris)
```

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)
```

```
from sklearn.naive_bayes import GaussianNB
model=GaussianNB()
model.fit(x_train, y_train)
y_pred=model.predict(x_test)
print (y_pred)
from sklearn.metrics import accuracy_score
print(f"accuracy={accuracy_score(y_test,y_pred)*100}%")
res=model.predict([[4.8,3.1,1.6,0.2]])
print(f"Result={iris.target_names[res[0]]}")
```

4. Result/Output:

A.



B.

```
[0 1 1 0 0 2 1 0 2 0 2 1 2 1 1 1 0 0 0 2 2 1 2 0 1 0 0 1 0 1]
accuracy=100.0%
Result=setosa
```

5. Learning outcomes (What I have learnt):

- Gained knowledge of **Decision Tree** as a supervised learning algorithm.
- Learned to apply Decision Tree for **classification of the Iris plant dataset**.
- Understood the working principle of **Naïve Bayes** as a probabilistic classifier.
- Learned to apply Gaussian Naïve Bayes for dataset classification.